SYED ARMAAN

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1.While Loop Example

```
count = 0
while (count < 3):
    count = count + 1
    print("Hello Good Morning")
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Hello Good Morning
Hello Good Morning
Hello Good Morning
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

2.For Loop Example

```
fruits = ["apple", "banana", "cherry"]

for fruit in fruits:

  print(fruit)
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
apple
banana
cherry
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

3. Simple Calculator (Functions)

```
def add(a, b):
  return a + b
def multiply(a, b):
 return a * b
x = int(input("Enter first number: "))
y = int(input("Enter second number: "))
print("Sum:", add(x, y))
print("Product:", multiply(x, y))
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter first number: 23
Enter second number: 43
Sum: 66
Product: 989
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

4. Temperature Checker (If-Else)

```
temp = int(input("Enter temperature: "))
if temp > 30:
 print("Hot")
elif temp > 20:
  print("Warm")
else:
  print("Cold")
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter temperature: 23
Warm
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter temperature: 33
Hot
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

5.Student Record (Dictionary)

```
students = {"Ali": 80, "Sara": 90}

name = input("Enter name: ")

if name in students:

 print("Marks:", students[name])

else:

print("Student not found")
```

```
Enter name: armaan
Student not found
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter name: ali
Student not found
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter name: Ali
Marks: 80
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

6.Tuple Example

```
coordinates = (4, 5)

print(coordinates[0])

print(coordinates[1])
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
4
5
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

7.List Input Example

movies = ["Leo", "Jawan", "RRR"]

print("Available movies:", movies)

choice = input("Enter movie to book: ")

if choice in movies:

print("Ticket booked for", choice)

movies.remove(choice)

else:

print("Movie not available")

```
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Available movies: ['Leo', 'Jawan', 'RRR']
Enter movie to book: alpha
Movie not available
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Available movies: ['Leo', 'Jawan', 'RRR']
Enter movie to book: Jawan
Ticket booked for Jawan
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

8. Classroom Attendance (Set)

students = set()

students.add("Ali")

students.add("John")

name = input("Enter name: ")

if name in students:

print("Present")

else:

print("Absent")

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter name: armaan
Absent
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

9.String Slicing

s = "ABCDEF"

print(s[1:4])

print(s[:3])

print(s[3:])

print(s[::-1])

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
BCD
ABC
DEF
FEDCBA
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```

10.Array

import array as arr

a = arr.array('i', [1, 2, 3])

print(a[0])

a.append(5)

print(a)

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
1
array('i', [1, 2, 3, 5])
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>
```